

0 3 . 1 Give **two** features of **all** prokaryotic cells that are **not** features of eukaryotic cells.

[1 mark]

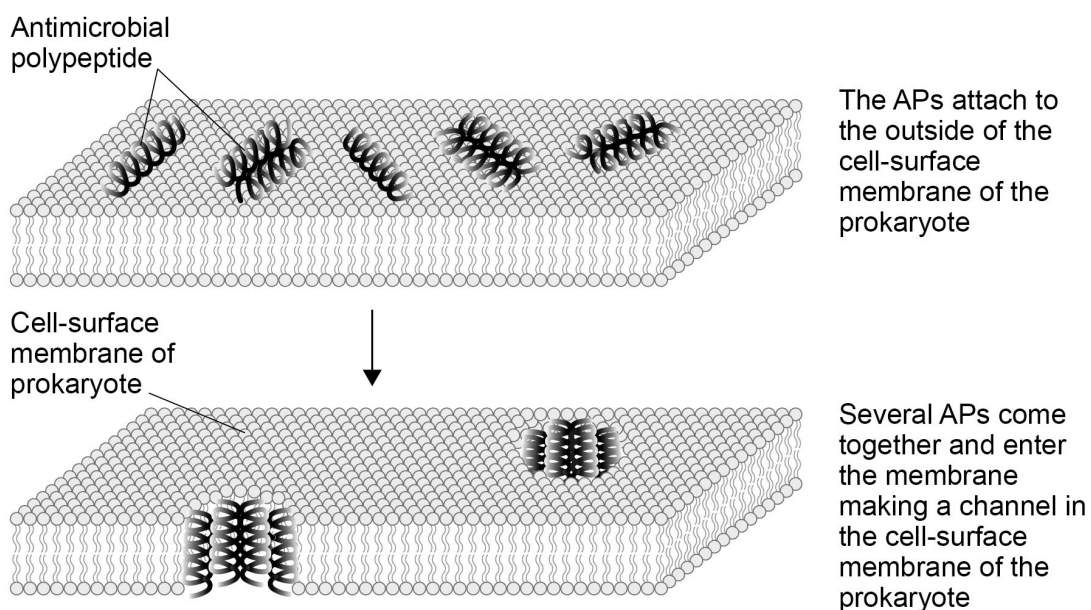
1 _____

2 _____

Many multicellular organisms produce antimicrobial polypeptides (APs) that protect them against prokaryotes.

Figure 3 shows how one type of AP acts on the cell-surface membrane of prokaryotes.

Figure 3



0 3 . 2 This AP has a secondary structure in a helical shape.

Tick (✓) the box to show which type of bond maintains the helical structure of the polypeptide.

[1 mark]

Disulfide

☐

Hydrogen

☐

Ionic

☐

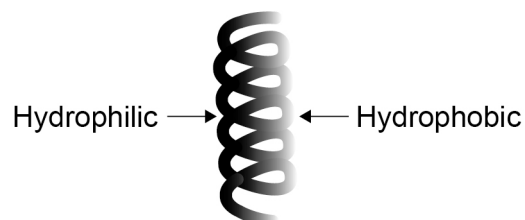
Peptide

☐

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The amino acids on one side of each AP helix have hydrophobic properties. The amino acids on the opposite side of each helix have hydrophilic properties. **Figure 4** shows this.

Figure 4

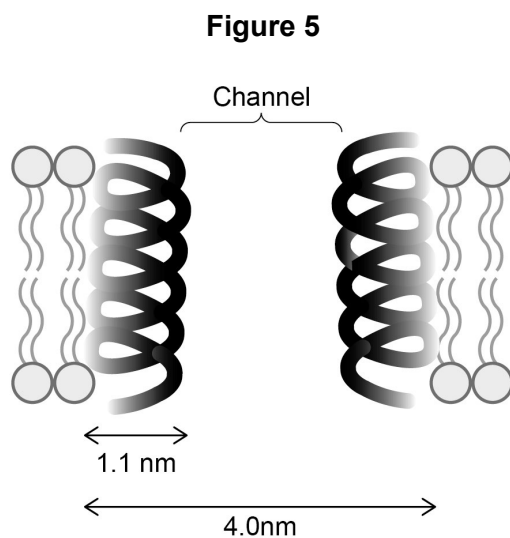


Suggest how these properties of the APs allow them to become positioned across the membrane (as shown in **Figure 3**) and make a channel through which ions can pass.

[2 marks]

Question 3 continues on the next page

Figure 5 shows further information about a channel formed in the cell-surface membrane by the APs.



0 3 . 4

Use **Figure 5** to calculate the cross-sectional area of the channel through which ions can pass.

Assume the cross-sectional area is circular.

Use $\pi = 3.14$ in your calculation. Give your answer in nm^2 **and** to 1 decimal place.

[2 marks]

Answer _____ nm^2

0 3 . 5

The APs damage prokaryotic cells but do not damage the eukaryotic cells in the organisms that produce them.
Prokaryotic cell membranes do not contain cholesterol.

Assess why the APs do not damage the eukaryotic cells of the organisms that produce them.

[2 marks]

0 3 . 6

Scientists observed these APs on prokaryotes using a transmission electron microscope. They stained the APs using a monoclonal antibody with gold attached to it.

Suggest how these techniques allowed observation of APs on prokaryotes.

[3 marks]
